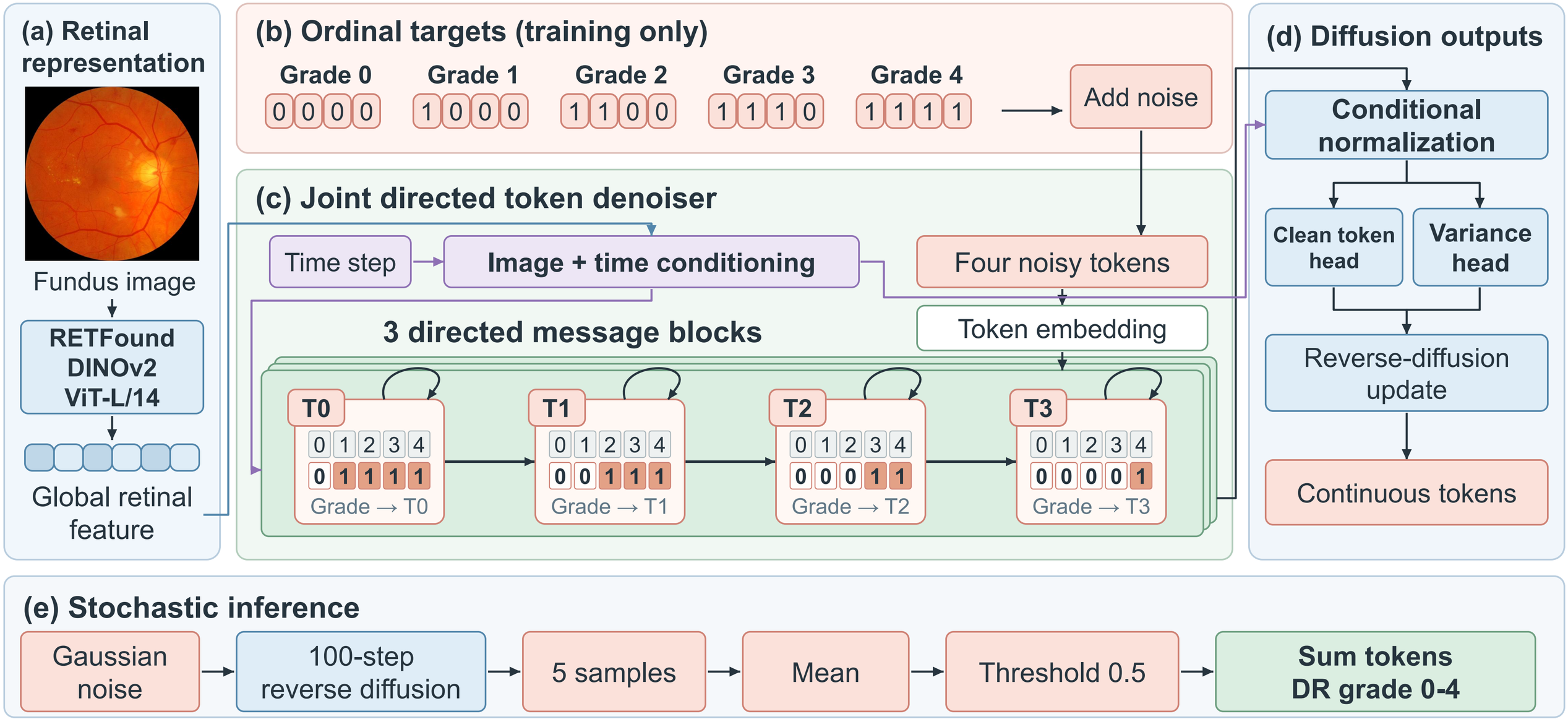

(a) Retinal representation
Fundus image
RETFound DINOv2 ViT-L/14
Global retinal feature
(b) Ordinal targets (training only)
Grade 0
0 0 0 0
Grade 1
1 0 0 0
Grade 2
1 1 0 0
Grade 3
1 1 1 0
Grade 4
1 1 1 1
Add noise
(c) Joint directed token denoiser
Time step
Image + time conditioning
Four noisy tokens
Token embedding
3 directed message blocks
T0
0 1 2 3 4
0 1 1 1 1
Grade → T0
T1
0 1 2 3 4
0 0 1 1 1
Grade → T1
T2
0 1 2 3 4
0 0 0 1 1
Grade → T2
T3
0 1 2 3 4
0 0 0 0 1
Grade → T3
(d) Diffusion outputs
Conditional normalization
Clean token head
Variance head
Reverse-diffusion update
Continuous tokens
(e) Stochastic inference
Gaussian noise
100-step reverse diffusion
5 samples
Mean
Threshold 0.5
Sum tokens DR grade 0-4